

Лабораторная работа № 6

Установка и настройка MySQL

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Цель:

Получить навыки установки, настройки и резервного копирования MariaDB/MySQL.



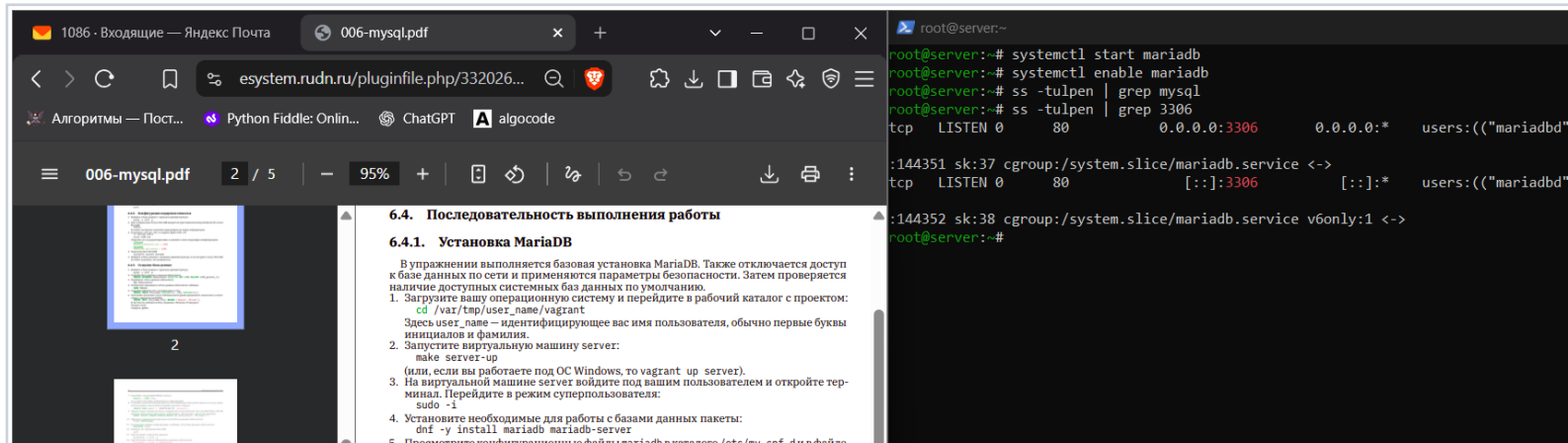
Установка MariaDB

```
root@server:~  
root@server:~# cat /etc/my.cnf  
#  
# This group is read both both by the client and the server  
# use it for options that affect everything  
#  
[client-server]  
#  
# include all files from the config directory  
#  
includedir /etc/my.cnf.d  
  
root@server:~# ls -la /etc/my.cnf.d  
total 56  
drwxr-xr-x. 2 root root 4096 Sep  5 21:38 .  
drwxr-xr-x. 148 root root 8192 Sep  5 21:38 ..  
-rw-r--r--. 1 root root  42 Jun 30 00:00 auth_gssapi.cnf  
-rw-r--r--. 1 root root 295 Jun 30 00:00 client.cnf  
-rw-r--r--. 1 root root 763 May 23 15:01 enable_encryption.preset  
-rw-r--r--. 1 root root 1460 Jun 30 00:00 mariadb-server.cnf  
-rw-r--r--. 1 root root 232 May 23 15:01 mysql-clients.cnf  
-rw-r--r--. 1 root root  76 May 23 15:01 provider_bzip2.cnf  
-rw-r--r--. 1 root root  72 May 23 15:01 provider_lz4.cnf  
-rw-r--r--. 1 root root  72 May 23 15:01 provider_lzo.cnf  
-rw-r--r--. 1 root root  78 May 23 15:01 provider_snappy.cnf  
-rw-r--r--. 1 root root 120 May 23 15:01 spider.cnf  
root@server:~# for f in /etc/my.cnf.d/*.cnf; do  
    echo "==== $f ====="  
    cat "$f"  
done  
==== /etc/my.cnf.d/auth_gssapi.cnf =====  
[mariadb]  
#plugin-load-add=auth_gssapi.so  
==== /etc/my.cnf.d/client.cnf =====  
#  
# These two groups are read by the client library  
# Use it for options that affect all clients, but not the server  
#  
[client]  
#  
# This group is not read by mysql client library,  
# If you use the same .cnf file for MySQL and MariaDB,  
# use it for MariaDB-only client options  
[client-mariadb]  
  
==== /etc/my.cnf.d/mariadb-server.cnf =====  
#  
# These groups are read by MariaDB server.  
# Use it for options that only the server (but not clients) should see  
#  
# See the examples of server my.cnf files in /usr/share/mysql/  
#
```

Смысл результата

Установлены серверные и клиентские пакеты MariaDB, служба запущена и включена. Статус systemd показывает успешную инициализацию СУБД.

Первичная защита СУБД



Смысл результата

Запущена процедура mariadb-secure-installation: удалены анонимные пользователи, тестовая база и небезопасные варианты удалённого доступа.

Проверка соединения

```
root@server:~# systemctl start mariadb
root@server:~# systemctl enable mariadb
root@server:~# ss -tulpen | grep mysql
root@server:~# ss -tulpen | grep 3306
tcp    LISTEN 0      80      0.0.0.0:3306      0.0.0.0:*    users:(("mariadb",pid=17268,fd=17))
                                     uid:27 ino
:144351 sk:37 cgroup:/system.slice/mariadb.service <->
tcp    LISTEN 0      80      [::]:3306      [::]:*    users:(("mariadb",pid=17268,fd=18))
                                     uid:27 ino
:144352 sk:38 cgroup:/system.slice/mariadb.service v6only:1 <->
root@server:~#
```

Смысл результата

Выполнено подключение к серверу и просмотрены версия MariaDB, набор символов и параметры службы. Это подтверждает готовность СУБД принимать SQL-команды.

Создание базы addressbook

```
root@server:~  
You already have your root account protected, so you can safely answer 'n'.  
  
Switch to unix_socket authentication [Y/n]  
Enabled successfully!  
Reloading privilege tables..  
... Success!  
  
You already have your root account protected, so you can safely answer 'n'.  
  
Change the root password? [Y/n] y  
New password:  
Re-enter new password:  
Password updated successfully!  
Reloading privilege tables..  
... Success!  
  
By default, a MariaDB installation has an anonymous user, allowing anyone  
to log into MariaDB without having to have a user account created for  
them. This is intended only for testing, and to make the installation  
go a bit smoother. You should remove them before moving into a  
production environment.  
  
Remove anonymous users? [Y/n] y  
... Success!  
  
Normally, root should only be allowed to connect from 'localhost'. This  
ensures that someone cannot guess at the root password from the network.  
  
Disallow root login remotely? [Y/n] y  
... Success!  
  
By default, MariaDB comes with a database named 'test' that anyone can  
access. This is also intended only for testing, and should be removed  
before moving into a production environment.  
  
Remove test database and access to it? [Y/n] y  
- Dropping test database...  
... Success!  
- Removing privileges on test database...  
... Success!  
  
Reloading the privilege tables will ensure that all changes made so far  
will take effect immediately.  
  
Reload privilege tables now? [Y/n] y  
... Success!  
  
Cleaning up...  
  
All done! If you've completed all of the above steps, your MariaDB  
installation should now be secure.
```

Смысл результата

Создана база данных addressbook и
выбрана для дальнейшей работы.
Команда SHOW DATABASES
подтверждает её наличие.

Создание таблицы city

```
root@server:~# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 13
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> \h

General information about MariaDB can be found at
http://mariadb.org

List of all client commands:
Note that all text commands must be first on line and end with ';'
?      (\?) Synonym for 'help'.
charset (\C) Switch to another charset. Might be needed for processing binlog with multi-byte charsets
clear  (\c) Clear the current input statement.
connect (\r) Reconnect to the server. Optional arguments are db and host.
delimiter (\d) Set statement delimiter.
edit    (\e) Edit command with $EDITOR.
ego     (\G) Send command to MariaDB server, display result vertically.
exit    (\q) Exit mysql. Same as quit.
go      (\g) Send command to MariaDB server.
help    (\h) Display this help.
nopager (\n) Disable pager, print to stdout.
notee   (\t) Don't write into outfile.
nowarning (\w) Don't show warnings after every statement.
pager   (\P) Set PAGER [to_pager]. Print the query results via PAGER.
print   (\p) Print current command.
prompt  (\R) Change your mysql prompt.
quit    (\q) Quit mysql.
rehash  (\#) Rebuild completion hash.
sandbox (\-) Disallow commands that access the file system (except \P without an argument and \e).
source  (\.) Execute an SQL script file. Takes a file name as an argument.
status  (\s) Get status information from the server.
system  (\!) Execute a system shell command.
tee      (\T) Set outfile [to_outfile]. Append everything into given outfile.
use      (\u) Use another database. Takes database name as argument.
warnings (\W) Show warnings after every statement.

For server side help, type 'help contents'

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
```

Смысл результата

В базе создана таблица city с ключевым полем и названием города. DESCRIBE показывает типы столбцов и первичный ключ.

Добавление тестовых данных

```
root@server:~  
root@server:~# mysql -u root -p  
Enter password:  
Welcome to the MariaDB monitor. Commands end with ; or \g.  
Your MariaDB connection id is 14  
Server version: 10.11.18-MariaDB MariaDB Server  
  
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.  
  
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.  
  
MariaDB [(none)]> status  
-----  
mysql Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using Editline wrapper  
  
Connection id:          14  
Current database:         
Current user:           root@localhost  
SSL:                    Not in use  
Current pager:          stdout  
Using outfile:             
Using delimiter:        ;  
Server:                 MariaDB  
Server version:         10.11.18-MariaDB MariaDB Server  
Protocol version:       10  
Connection:             Localhost via UNIX socket  
Server characterset:    latin1  
Db characterset:        latin1  
Client characterset:    utf8mb3  
Conn. characterset:     utf8mb3  
UNIX socket:            /var/lib/mysql/mysql.sock  
Uptime:                 5 min 49 sec  
  
Threads: 1 Questions: 27 Slow queries: 0 Opens: 20 Open tables: 13 Queries per second avg: 0.077  
-----  
  
MariaDB [(none)]> exit  
Bye  
root@server:~#
```

Смысл результата

В таблицу внесены несколько городов, после чего SELECT вывел сохранённые строки. Так проверена запись и чтение данных.

Изменение и удаление строк

```
root@server:/etc/my.cnf.d
root@server:~# cd /etc/my.cnf.d
root@server:/etc/my.cnf.d# touch utf8.cnf
root@server:/etc/my.cnf.d# nano utf.cnf
root@server:/etc/my.cnf.d# root@server:/etc/my.cnf.d#
root@server:/etc/my.cnf.d# nano utf8.cnf
root@server:/etc/my.cnf.d# systemctl restart mariadb
root@server:/etc/my.cnf.d# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> status
-----
mysql Ver 15.1 Distrib 10.11.18-MariaDB, for Linux (x86_64) using Editline wrapper

Connection id:          3
Current database:
Current user:            root@localhost
SSL:                    Not in use
Current pager:           stdout
Using outfile:           ''
Using delimiter:         ;
Server:                  MariaDB
Server version:          10.11.18-MariaDB MariaDB Server
Protocol version:        10
Connection:              Localhost via UNIX socket
Server characterset:      utf8mb3
Db characterset:          utf8mb3
Client characterset:      utf8mb3
Conn. characterset:       utf8mb3
UNIX socket:              /var/lib/mysql/mysql.sock
Uptime:                  20 sec

Threads: 1  Questions: 4  Slow queries: 0  Opens: 17  Open tables: 10  Queries per second avg: 0.200
-----

MariaDB [(none)]>
```

Смысл результата

Команды UPDATE и DELETE изменили тестовый набор. Повторный SELECT позволяет увидеть фактический результат каждой операции.

Отдельный пользователь БД

```
root@server:/etc/my.cnf.d
root@server:/etc/my.cnf.d# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 4
Server version: 10.11.18-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE addressbook CHARACTER SET utf8 COLLATE utf8_general_ci;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> USE addressbook;
Database changed
MariaDB [addressbook]> SHOW TABLES;
Empty set (0.001 sec)

MariaDB [addressbook]> CREATE TABLE city(name VARCHAR(40), city VARCHAR(40));
Query OK, 0 rows affected (0.026 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Иванов','Москва');
Query OK, 1 row affected (0.004 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Петров','Сочи');
Query OK, 1 row affected (0.003 sec)

MariaDB [addressbook]> INSERT INTO city(name,city) VALUES ('Сидоров','Дубна');
Query OK, 1 row affected (0.005 sec)

MariaDB [addressbook]> SELECT * FROM city;
+-----+-----+
| name | city |
+-----+-----+
| Иванов | Москва |
| Петров | Сочи |
| Сидоров | Дубна |
+-----+-----+
rows in set (0.001 sec)

MariaDB [addressbook]>
```

Смысл результата

Создан пользователь приложения и назначены права только на базу addressbook. Такой подход отделяет административную учётную запись от прикладного доступа.

Проверка выданных привилегий

```
root@server:/etc/my.cnf.d
MariaDB [addressbook]> CREATE USER 'ndbaranov'@'%' IDENTIFIED BY 'password';
Query OK, 0 rows affected (0.003 sec)

MariaDB [addressbook]> GRANT SELECT,INSERT,UPDATE,DELETE
    -> ON addressbook.*
    -> TO 'ndbaranov'@'%';
Query OK, 0 rows affected (0.003 sec)

MariaDB [addressbook]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [addressbook]> DESCRIBE city;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| name  | varchar(40)   | YES  |     | NULL    |       |
| city  | varchar(40)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.002 sec)

MariaDB [addressbook]> quit
Bye
root@server:/etc/my.cnf.d# mysqlshow -u root -p
Enter password:
+-----+
| Databases |
+-----+
| addressbook |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
root@server:/etc/my.cnf.d# mysqlshow -u root -p addressbook
Enter password:
Database: addressbook
+-----+
| Tables |
+-----+
| city |
+-----+
```

Смысл результата

Под новой учётной записью выполнены разрешённые операции с таблицей. SHOW GRANTS фиксирует ограниченный набор полномочий.

Логическая резервная копия

```
Выбрать root@server:~
root@server:/etc/my.cnf.d# cd
root@server:~# mkdir -p /var/backup
root@server:~# mysqldump -u root -p addressbook > /var/backup/addressbook.sql
Enter password:
root@server:~# mysqldump -u root -p addressbook | gzip > /var/backup/addressbook.sql.gz
Enter password:
root@server:~# mysqldump -u root -p addressbook | gzip > $(date
+ /var/backup/addressbook.%Y%m%d.%H%M%S.sql.gz)
Enter password: bash: +: command not found...
-bash: $(date
+ /var/backup/addressbook.%Y%m%d.%H%M%S.sql.gz): ambiguous redirect
mysqldump: Got errno 32 on write
root@server:~# mysql -u root -p addressbook < /var/backup/addressbook.sql
Enter password:
root@server:~# zcat /var/backup/addressbook.sql.gz | mysql -u root -p addressbook
Enter password:
root@server:~#
```

Смысл результата

Утилита mysqldump выгрузила структуру и данные addressbook в SQL-файл. Размер и начало файла подтверждают наличие команд восстановления.

Сжатие и проверка копии

```
PowerShell 7 (x64)
PS D:\work\ndbaranov\vagrant> cd D:\work\ndbaranov\vagrant
PS D:\work\ndbaranov\vagrant> notepad .\provision\server\mysql.sh
PS D:\work\ndbaranov\vagrant> vagrant validate
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
Vagrantfile validated successfully.
PS D:\work\ndbaranov\vagrant> vagrant provision server --provision-with "server mysql"
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Running provisioner: server mysql (shell)...
server: Running: C:/Users/barash/AppData/Local/Temp/vagrant-shell120260906-3112-tsbphu.sh
server: Provisioning MariaDB server
server: Last metadata expiration check: 3:00:24 ago on Sat 05 Sep 2026 06:57:05 PM UTC.
server: Package mariadb-3:10.11.18-1.el10_2.x86_64 is already installed.
server: Package mariadb-server-3:10.11.18-1.el10_2.x86_64 is already installed.
server: Dependencies resolved.
server: Nothing to do.
server: Complete!
server: MariaDB provisioning completed
PS D:\work\ndbaranov\vagrant> █
```

Смысл результата

Дамп сжат, затем его содержимое проверено без ручного редактирования. Provision-сценарий завершает воспроизводимую настройку MariaDB.

Результат

MariaDB защищена базовыми настройками, создана база addressbook и пользователь с ограниченными правами. Логическая резервная копия позволяет восстановить структуру и данные.